## a Coding-invariant design

| | C | C | H | H |
|---|---|---|---|---|
| C | 0 | 0 | 1 | 1 |
| C | 0 | 0 | 1 | 1 |
| H | 1 | 1 | 0 | 0 |
| H | 1 | 1 | 0 | 0 |

0 = same group; 1 = different group

Independent of numeric label coding

## b Thirty simulated axis cases

| | No effect | Phase mod. | Shape change | Strong axis |
|---|---|---|---|---|
| Inert | 6 | 0 | 0 | 0 |
| Common phase | 0 | 6 | 0 | 0 |
| Gene-specific | 0 | 0 | 6 | 0 |
| Independent phase | 0 | 0 | 0 | 6 |
| Discrete shift | 0 | 0 | 0 | 6 |

Six seeded cases per scenario

## c Real phase diagnostics

Shares describe circular phase dispersion, not total expression variance.